

VueJS - Components

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Ex2: Vue Application – Containing multiple components

Header.vue

`<script setup>` Create a reactive variable using ref. Vue automatically detects the change and updates the HTML wherever emoji is used

```
import { ref } from 'vue'
```

```
const emoji = ref('👌')
```

```
</script>
```

```
<template>
```

```
  <header>
```

In Vue.js, {{ }} is called interpolation (or text interpolation). It is used to display a JavaScript value inside the HTML template.

```
    <span>{{ emoji }} Praba</span>
```

```
  </header>
```

```
</template>
```

```
<style scoped>
```

```
  header span {
```

```
    font-size: 3rem;
```

```
}
```

```
</style>
```

Output →



Praba

App.vue

```
<script setup>
```

```
import Header from './components/Header.vue'
```

```
</script>
```

```
<template>
```

```
  <Header/>
```

```
</template>
```

3rem is a CSS length unit.

By default, browsers usually use 16px

Therefore: $3\text{rem} = 3 \times 16\text{px} = 48\text{px}$

v 1.2

Main.vue

```
<script setup>
import {ref} from 'vue'
const name=ref('Prabavathy')
</script>
<template>
<main>
  <h1>Hello {{ name }}!</h1>
  <p>I am about to learn <a
href="https://vuejs.org/"
target="_blank">Vue.js</a></p>
</main>
</template>
<style scoped>
main h1{
  display:block;
}
</style>
```

Ex2: Vue Application – Containing multiple components

Hello World Praba



Hello Prabavathy!

I am about to learn **Vue.js**

Output →

App.vue

```
<script setup>
import HelloWorld from './components/HelloWorld.vue'
import Header from './components/Header.vue'
import Main from './components/Main.vue'
</script>
<template>
  <HelloWorld/>
  <Header/>
  <Main/>
</template>
```

v 1.2

Footer.vue

```
<script setup>
import {ref} from 'vue'
const year=ref(2026)
</script>
<template>
  <p>&copy; {{ year }}</p>
</template>
<style scoped>
p{
  position: fixed;
  bottom:0;
  padding-bottom: 20px;
}
</style>
```

Ex2: Vue Application – Containing multiple components

Output ➔

Hello World Praba



Hello Prabavathy!

I am about to learn [Vue.js](#)

© 2026

App.vue

```
<script setup>
import HelloWorld from './components/HelloWorld.vue'
import Header from './components/Header.vue'
import Main from './components/Main.vue'
import Footer from './components/Footer.vue';
</script>
<template>
  <HelloWorld/>
  <Header/>
  <Main/>
  <Footer/>
</template>
```

Student.vue

```
<script setup>
defineProps({
  name: String,
  department: String
})
```

```
</script>
```

```
<template>
```

```
  <div>
```

```
    <h2>{{ name }}</h2>
```

```
    <p>Department: {{ department }}</p>
```

```
  </div>
```

```
</template>
```

```
<style scoped>
```

```
div p{
  color:blue;
}
```

```
</style>
```

The component expects two input values (props) from its parent component: name and department, and both should be strings.

Output →

Student Details

Prabavathy

Department: CSE

App.vue

```
<script setup>
import Student from
'./components/Student.vue'
```

```
</script>
```

```
<template>
```

```
  <h1>Student Details</h1>
```

```
  <Student name="Prabavathy"
department="CSE" />
```

```
</template>
```

```
<style scoped>
```

```
</style>
```

In Vue.js, an alias in importing means giving a short or alternative name/path to a file or folder so that you don't have to write the complete path every time.

Example without alias

Suppose your component is located at:

src/

```
|— components/
|   |— Header.vue
|   |— App.vue
```

You can import it using:

```
import Header from
'./components/Header.vue'
```

Here, ./components/Header.vue is the relative path.

Using an alias

Vue projects commonly configure @ as an alias for the src folder.

So instead of:

```
import Header from
'./components/Header.vue'
you can write:
import Header from
'@/components/Header.vue'
```

Here:

@ → src

So:

@/components/Header.vue
means:
src/components/Header.vue

Ex4: Develop a **Student Profile** page. You want to display:

- Student name
- Department
- Student photo
- Whether the student is active
- List of courses
- A button to increase the student's completed course count

Student Details

Anitha

Department: Computer Science



Student Photo

Status: Active

Courses

- Vue.js
- Java
- Python

Completed Courses: 2

Complete Course



```
<script setup>
import { ref } from 'vue'
const studentName = ref('Anitha')
const department = ref('Computer Science')
const photo =
ref('https://example.com/student.jpg')
const isActive = ref(true)
const courses = ref(['Vue.js', 'Java', 'Python'])
const completedCourses = ref(2)
</script>
```

```
<style scoped>
.student {
  width: 400px;
  padding: 20px;
  border: 1px solid gray;
}
```

```
img {
  width: 120px;
}
```

```
button {
  padding: 8px 15px;
}
</style>
```

```
<template>
  <div class="student">
    <!-- 1. Interpolation -->
    <h1>{{ studentName }}</h1>
    <p>Department: {{ department }}</p>
    <!-- 2. Attribute binding -->
    
    <!-- 3. Conditional rendering -->
    <p v-if="isActive">Status: Active</p>
    <p v-else>Status: Inactive</p>
    <!-- 4. List rendering -->
    <h3>Courses</h3>
    <ul>
      <li v-for="course in courses" :key="course">
        {{ course }}
      </li>
    </ul>
    <!-- 5. Event binding -->
    <p>Completed Courses: {{ completedCourses
  }}</p>
    <button @click="completedCourses++">
      Complete Course
    </button>
  </div>
</template>
```